

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO3: *Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)*

Assessment Tool Weightage: 40%

Duration :1 Hour

QUESTIONS: 4

Total Marks:40

Annexure A

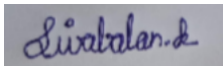
CASE STUDY

Code of Conduct:

I Siva Balan K (192572037) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Annexure A

CASE STUDY

Case Study 1 — Smart Medical Diagnosis System

A hospital is designing a rule-based Logical Agent for preliminary patient diagnosis. The system's knowledge base contains:

- I. If a patient has Fever AND Cough $\rightarrow$ Possible Flu
- II. If a patient has Fever AND Rash $\rightarrow$ Possible Measles
- III. If a patient has Cough AND Breathlessness $\rightarrow$ Possible Pneumonia
- IV. Facts about Patient A: Fever = True, Cough = True, Breathlessness = True

- (a) Represent the above rules and facts precisely using Propositional Logic symbols. Define each symbol used.
- (b) Apply Forward Chaining on the knowledge base to derive all possible diagnoses for Patient A. Show each inference step with the rule fired and fact derived.
- (c) The goal is to confirm 'Patient A has Pneumonia.' Apply Backward Chaining and draw the goal reduction tree to verify this goal.

Case Study 2 — Autonomous Traffic Management Agent

A smart city traffic controller uses a FOL-based knowledge base to manage emergency vehicle routing:

- I. $\forall x: \text{Vehicle}(x) \wedge \text{EmergencyType}(x) \rightarrow \text{ClearPath}(x)$ [Rule 1]
- II. $\forall x: \text{ClearPath}(x) \rightarrow \text{GreenSignal}(x) \wedge \neg \text{RedSignal}(x)$ [Rule 2]
- III. $\forall x \forall y: \text{Vehicle}(x) \wedge \text{Behind}(x,y) \wedge \text{GreenSignal}(y) \rightarrow \text{Proceed}(x)$ [Rule 3]
- IV. Facts: Vehicle(Ambulance), EmergencyType(Ambulance), Vehicle(CarA), Behind(CarA, Ambulance)

- (a) Apply Unification to match Rule 1 with the given facts. Show the complete substitution set θ (MGU) at each step.
- (b) Using the Resolution method, prove that 'Proceed (CarA)' is true. Convert relevant clauses to CNF and show each resolution step.
- (c) Analyze whether the current FOL knowledge base is sufficient to handle two simultaneous emergency vehicles. Identify what additional rules or facts are needed and justify logically.

Case Study 3 — Agricultural Expert Reasoning System

An agricultural advisory logical agent operates with the following propositional knowledge base (KB):

- 1) P1: SoilDry $\rightarrow$ IrrigationNeeded
- 2) P2: IrrigationNeeded $\wedge$ CropWheat $\rightarrow$ ApplyDripMethod
- 3) P3: $\neg \text{ApplyDripMethod} \wedge \text{CropWheat} \rightarrow \text{CropAtRisk}$
- 4) Facts: SoilDry = True, CropWheat = True

- (a) Convert all sentences P1, P2, P3 and the facts into Conjunctive Normal Form (CNF). Show each conversion step (eliminate implication $\rightarrow$ move $\neg$ inward $\rightarrow$ distribute).

(b) Using Resolution Refutation, prove the theorem 'ApplyDripMethod'. Negate the goal, combine with the KB in CNF, and derive the empty clause $\square$. Draw the resolution proof tree.

(c) If the fact 'SoilDry' is removed from the KB, analyze step-by-step how logical conclusions change. Does 'CropAtRisk' still follow? Apply appropriate reasoning patterns to justify.

Case Study 4 — Wumpus World Knowledge Agent

An AI agent navigating the Wumpus World perceives: Stench at [1,2], Breeze at [1,1], Glitter at [2,2]. The agent uses a propositional KB with these rules:

- 1) Stench [1,2] $\rightarrow$ Wumpus is adjacent to cell [1,2]
- 2) Breeze [1,1] $\rightarrow$ Pit is adjacent to cell [1,1]
- 3) Glitter [x,y] $\rightarrow$ Gold is at [x,y]
- 4) $\neg$ Wumpus [x,y] $\wedge$ $\neg$ Pit[x,y] $\rightarrow$ Safe[x,y]

(a) Construct the agent's belief state from its percepts. Apply Modus Ponens systematically to derive new knowledge facts. List all derived conclusions.

(b) Using Forward Chaining from the agent's percepts, identify the possible locations of the Wumpus and Pit. Show each reasoning step precisely.

(c) Evaluate whether the path [1,1] $\rightarrow$ [1,2] $\rightarrow$ [2,2] is safe for the agent to collect the gold. Use the knowledge derived in (a) and (b) to justify your conclusion with logical inference.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding ; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided

		solutions with strong justification			
Clarity	10	Clear, wellstructured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

CASE STUDY 1 — Smart Medical Diagnosis System

(a) Propositional Logic Representation

Let:

- **F** = Patient has Fever
- **C** = Patient has Cough
- **R** = Patient has Rash
- **B** = Patient has Breathlessness
- **Flu** = Possible Flu
- **M** = Possible Measles
- **P** = Possible Pneumonia

Rules

1. Fever AND Cough $\rightarrow$ Possible Flu
 $F \wedge C \rightarrow \text{Flu}$
2. Fever AND Rash $\rightarrow$ Possible Measles
 $F \wedge R \rightarrow M$
3. Cough AND Breathlessness $\rightarrow$ Possible Pneumonia
 $C \wedge B \rightarrow P$

Facts

F = True, C = True, B = True

(b) Forward Chaining

Initial facts:

F, C, B

Step 1

Rule 1:

$F \wedge C \rightarrow \text{Flu}$

Since F and C are true:

Flu = True

Step 2

Rule 3:

$C \wedge B \rightarrow P$

Since C and B are true:

Pneumonia = True

Step 3

Rule 2 requires **$F \wedge R$** , but Rash is not given.

Therefore:

Measles cannot be derived.

Result

The possible diagnoses are:

Possible Flu

Possible Pneumonia

(c) Backward Chaining for Pneumonia

Goal:

Pneumonia

Find a rule whose conclusion is Pneumonia:

$C \wedge B \rightarrow \text{Pneumonia}$

Therefore, the goal is reduced to:

$\text{Pneumonia} \leftarrow \text{Cough} \wedge \text{Breathlessness}$

Facts:

Cough = True

Breathlessness = True

Both sub-goals are satisfied.

Goal Reduction Tree

Pneumonia

|

Cough AND Breathlessness

/ \

Cough=True Breathlessness=True

Result

Therefore:

Patient A has Possible Pneumonia.

CASE STUDY 2 — Autonomous Traffic Management Agent

(a) Unification

Rule 1:

$\forall x [\text{Vehicle}(x) \wedge \text{EmergencyType}(x) \rightarrow \text{ClearPath}(x)]$

Given:

Vehicle(Ambulance)

EmergencyType(Ambulance)

Match:

Vehicle(x) = Vehicle(Ambulance)

EmergencyType(x) = EmergencyType(Ambulance)

Therefore:

$\theta = \{x / \text{Ambulance}\}$

After substitution:

Vehicle(Ambulance) $\wedge$ EmergencyType(Ambulance) $\rightarrow$ ClearPath(Ambulance)

Hence:

ClearPath(Ambulance)

(b) Resolution Method

CNF Clauses

Rule 1:

$\neg \text{Vehicle}(x) \vee \neg \text{EmergencyType}(x) \vee \text{ClearPath}(x)$

Rule 2:

$\neg \text{ClearPath}(x) \vee \text{GreenSignal}(x)$

$\neg \text{ClearPath}(x) \vee \neg \text{RedSignal}(x)$

Rule 3:

$\neg \text{Vehicle}(x) \vee \neg \text{Behind}(x,y) \vee \neg \text{GreenSignal}(y) \vee \text{Proceed}(x)$

Facts:

Vehicle(Ambulance)

EmergencyType(Ambulance)

Vehicle(CarA)

Behind(CarA,Ambulance)

Resolution Steps

From Rule 1 and the facts:

Vehicle(Ambulance)

EmergencyType(Ambulance)

ClearPath(Ambulance)

From Rule 2:

ClearPath(Ambulance)

GreenSignal(Ambulance)

Now Rule 3 with:

x = CarA, y = Ambulance

Vehicle(CarA)

Behind(CarA,Ambulance)

GreenSignal(Ambulance)

Proceed(CarA)

Result

Proceed(CarA) is logically entailed by the KB.

(c) Two Simultaneous Emergency Vehicles

The current KB can handle multiple emergency vehicles because the rules use $\forall x$.

For example, if we have:

Vehicle(Ambulance1)

EmergencyType(Ambulance1)

Vehicle(Ambulance2)

EmergencyType(Ambulance2)

Rule 1 derives:

ClearPath(Ambulance1)

ClearPath(Ambulance2)

However, the KB does not contain rules for **conflicting emergency routes** or **priority between two emergency vehicles**.

Additional rules needed

For example:

Priority(Ambulance1, Ambulance2) $\rightarrow$ GivePriority(Ambulance1)

and facts describing priority, route conflicts, or locations.

Result

The KB can identify multiple emergencies, but additional **priority and conflict-resolution rules** are required for safe simultaneous routing.

CASE STUDY 3 — Agricultural Expert Reasoning System

(a) CNF Conversion

P1

SoilDry $\rightarrow$ IrrigationNeeded

Eliminate implication:

$\neg$ SoilDry $\vee$ IrrigationNeeded

This is already CNF.

P2

$\text{IrrigationNeeded} \wedge \text{CropWheat} \rightarrow \text{ApplyDripMethod}$

Eliminate implication:

$\neg(\text{IrrigationNeeded} \wedge \text{CropWheat}) \vee \text{ApplyDripMethod}$

Using De Morgan's law:

$\neg\text{IrrigationNeeded} \vee \neg\text{CropWheat} \vee \text{ApplyDripMethod}$

CNF:

$\neg I \vee \neg W \vee D$

P3

$\neg\text{ApplyDripMethod} \wedge \text{CropWheat} \rightarrow \text{CropAtRisk}$

Eliminate implication:

$\neg(\neg\text{ApplyDripMethod} \wedge \text{CropWheat}) \vee \text{CropAtRisk}$

Therefore:

$\text{ApplyDripMethod} \vee \neg\text{CropWheat} \vee \text{CropAtRisk}$

CNF:

$D \vee \neg W \vee R$

Facts

SoilDry

CropWheat

(b) Resolution Refutation

Goal

Prove:

ApplyDripMethod (D)

Negate the goal:

$\neg D$

Derivation

From:

$\neg \text{SoilDry} \vee \text{IrrigationNeeded}$

and:

SoilDry

we derive:

IrrigationNeeded

Now use:

$\neg \text{IrrigationNeeded} \vee \neg \text{CropWheat} \vee \text{D}$

Together with:

IrrigationNeeded

and:

CropWheat

we derive:

D

But the negated goal gives:

$\neg \text{D}$

Therefore:

D

$\neg \text{D}$

——

□

Resolution Proof Tree

SoilDry

|

+---- $\neg \text{SoilDry} \vee \text{IrrigationNeeded}$

|

IrrigationNeeded

$\neg I \vee \neg W \vee D$
+ W
D
$\neg D$
□

Result

Since the empty clause □ is derived:

ApplyDripMethod is proved.

(c) Removing SoilDry

If **SoilDry** is removed:

1. **IrrigationNeeded** cannot be derived from P1.
2. Without **IrrigationNeeded**, P2 cannot derive **ApplyDripMethod**.
3. **CropWheat** is still known.
4. However, P3 requires $\neg$ **ApplyDripMethod**.

The absence of **ApplyDripMethod** does **not** mean:

$\neg$ **ApplyDripMethod**

Therefore:

CropAtRisk cannot be logically concluded.

Result

Without **SoilDry**:

ApplyDripMethod — Not derivable

CropAtRisk — Not derivable

CASE STUDY 4 — Wumpus World Knowledge Agent

(a) Belief State and Modus Ponens

Percepts

Stench[1,2]

Breeze[1,1]

Glitter[2,2]

Step 1 — Stench

Rule:

Stench[1,2] $\rightarrow$ Wumpus adjacent to [1,2]

Therefore possible Wumpus locations are:

[1,1], [1,3], [2,2]

Step 2 — Breeze

Rule:

Breeze[1,1] $\rightarrow$ Pit adjacent to [1,1]

Therefore possible Pit locations are:

[1,2], [2,1]

Step 3 — Glitter

Rule:

Glitter[x,y] $\rightarrow$ Gold[x,y]

Therefore:

Gold[2,2]

Derived Knowledge

Wumpus $\in \{[1,1], [1,3], [2,2]\}$

Pit $\in \{[1,2], [2,1]\}$

Gold[2,2]

(b) Forward Chaining

Wumpus

From:

Stench[1,2]

derive:

Wumpus adjacent to [1,2]

↓

[1,1], [1,3], [2,2] are possible

Pit

From:

Breeze[1,1]

derive:

Pit adjacent to [1,1]

↓

[1,2], [2,1] are possible

Gold

From:

Glitter[2,2]

derive:

Gold[2,2]

(c) Safety of the Path

Proposed path:

[1,1] → [1,2] → [2,2]

From the knowledge base:

- **[1,2]** is a possible Pit location.
- **[1,2]** is also adjacent to the detected Stench.
- **[2,2]** is a possible Wumpus location.

- No facts establish $\neg\mathbf{Wumpus}$ or $\neg\mathbf{Pit}$ for these cells.

Rule 4 requires:

$$\neg\mathbf{Wumpus}[x,y] \wedge \neg\mathbf{Pit}[x,y] \rightarrow \mathbf{Safe}[x,y]$$

These negative facts are not available.

Result

The path **cannot be confirmed as safe**.

Therefore, the agent should **not assume that the path is safe** without additional percepts or knowledge.